

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #63 – Community Solar Credit Allocation Manager

USE-CASE FLOW SPECIFICATION

UC-03: Transfer Solar Credits

Primary Actor: Prosumer Resident

1. Preconditions

1. The prosumer resident is authenticated.
2. The prosumer has an active solar credit account.
3. The recipient is a valid participating resident.
4. The prosumer has available solar credits.

2. Postconditions

1. The transferred credits are deducted from the sender's credit balance.
2. The transferred credits are added to the recipient's credit balance.
3. The transfer transaction is recorded.
4. The updated credit balance is available for viewing.

3. Main Success Scenario

1. The prosumer selects **Transfer Solar Credits**.
2. The system displays the recipient account and credit amount fields.
3. The prosumer enters the recipient account and the number of credits to transfer.
4. The system validates the recipient account and transfer amount.
5. The system validates the prosumer's available credit balance.
6. The system confirms that the requested transfer does not exceed the available balance.
7. The system deducts the transferred credits from the prosumer's account.
8. The system adds the transferred credits to the recipient's account.

9. The system records the credit-transfer transaction.

10. The system displays a successful transfer confirmation.

4. Alternate Flow – Insufficient Credit Balance

5a. The system determines that the requested transfer exceeds the prosumer's available credit balance.

5a.1. The system rejects the transfer.

5a.2. The system displays "**Insufficient solar credit balance.**"

5a.3. The prosumer enters a lower transfer amount.

5a.4. No credits are deducted and no transaction is recorded.